CALMPA

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Make rest visible through calm physical interaction.

通过平静的物理交互，让休息变得可见

Calmpa is a desktop interactive device designed for office workers. Instead of alarms or pop-ups, it uses slow petal movement to gently reveal sedentary time. It turns sitting time into a visible physical change.

Calmpa是一款专为办公室工作人员设计的桌面互动设备。它不使用闹钟或弹出窗口，而是通过缓慢的花瓣运动来温和地提醒用户久坐时间，它将久坐时间转化为可见的物理变化。

Design Strategy 设计策略

Do Not Disturb Reminder 非打扰提醒

Reminds without interrupting encouraging natural breaks.

提醒而不打断，鼓励自然休息

Calm Physical Feedback 平静反馈

Slow, petal-like movements make sedentary change visible.

缓慢而如花瓣般的运动使久坐不动的变化变得可见

Desktop Integration 桌面整合

Compact, stable, and visually harmonious with the workspace.

结构紧凑、稳定，并且在视觉上与工作空间和谐统一

① **Calm Technology 平静技术** — Restrained movements and lighting blend into the everyday environment / 克制的动作与光线，融入日常环境

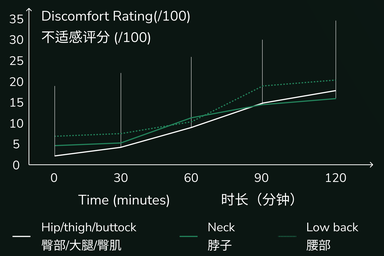
② **Sedentary Perception 久坐感知** — Based on a time-sensing mechanism, it gently records sedentary states / 基于时间的感知机制，温和记录久坐状态

③ **Privacy-Friendly 隐私友好** — Proximity sensing, no personal data, respecting user privacy / 不采集个人信息，仅通过接近感知用户状态

Background 背景问题

Prolonged desk work often makes sedentary time invisible. During focused tasks, users may overlook physical fatigue and delay breaks.

长时间伏案工作会让久坐时间变得不易察觉。在高专注任务中，用户容易忽视身体疲劳，并不断推迟休息。



Discomfort in the neck, lower back, and hip/thigh areas increases during prolonged sitting, showing that sedentary fatigue builds up and is easily overlooked.

长时间久坐会使颈部、腰部和臀部/大腿区域的不适感上升，说明久坐疲劳会逐渐累积，并容易被忽视。

数字提醒 Digital Reminders

- Wearable 可穿戴设备
- Pop-Up 弹窗提醒
- Alarm 手机警告

Digital reminders use alarms, pop-ups, apps, or wearables to prompt users to move at scheduled intervals.

数字提醒通过闹钟、弹窗、应用或可穿戴设备，定时提醒用户活动。

办公环境 Workplace Setup

- Meeting 会议转换
- Stand Desk 站立办公
- Office Layout 办公布局

Environmental interventions use workspace design to encourage movement.

环境干预通过办公桌、步行会议或空间布局，鼓励用户在工作中活动。

个人习惯 Personal Habits

- Planned Breaks 规划式休憩
- Stretching 伸展运动
- Time Plan 定时活动

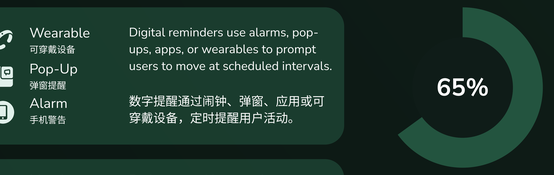
Habit strategies use breaks, stretching, or time planning to reduce prolonged sitting.

习惯策略通过休息、伸展或时间规划来减少长时间久坐。

User Insight 用户洞察

The problem is not the absence of reminders, but the lack of gentle awareness that helps focused users notice gradual physical fatigue.

问题不在于缺少提醒，而在于缺少一种温和的觉察方式，帮助专注中的用户意识到逐渐累积的身体疲劳。



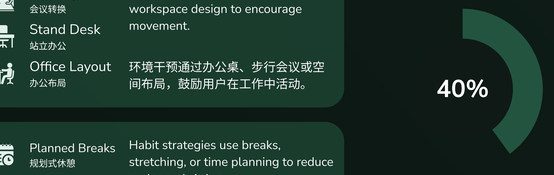
65% Office workers sit for more than 6 hours per day

65%的职场人群每日久坐超过6小时



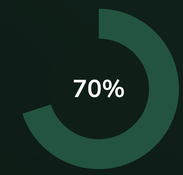
80% Average daily sitting time of whole working

职场人群平均每日久坐时间占总比80%



40% Over 8 hours of sitting increases mortality risk

久坐超过8小时，死亡风险增加40%



70% Users ignore or delay digital reminders

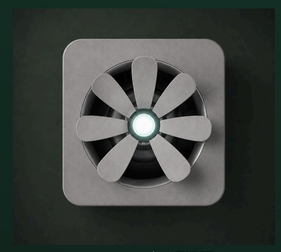
70%的用户会忽略或延迟数字提醒

Research Question 研究问题

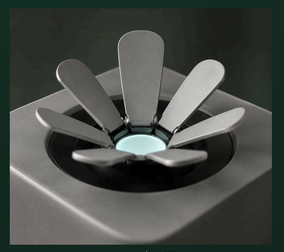
How can we make prolonged sitting more noticeable and encourage people to take breaks through peaceful interactions?

如何通过平和的互动使长时间久坐变得显而易见并鼓励人们休息？

Detailed Display 细节展示



Top View | 顶视图



Detail | 细节

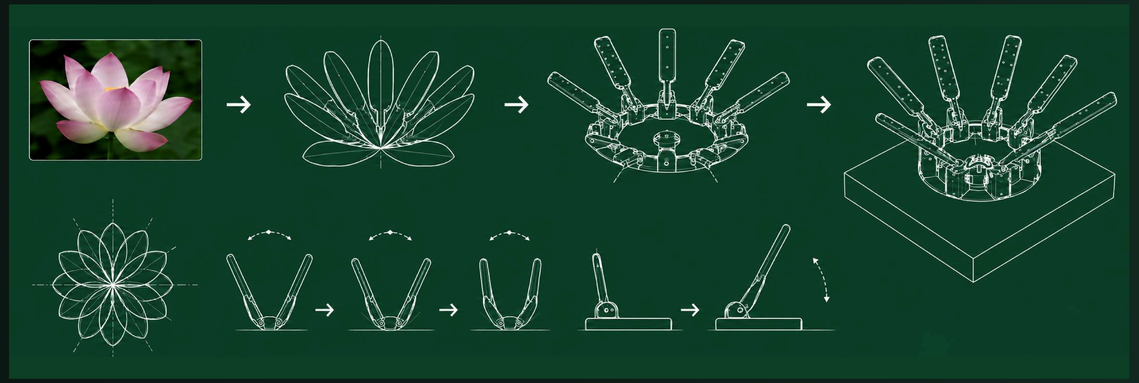


Desktop View | 桌面视图



Light Change | 光效变化

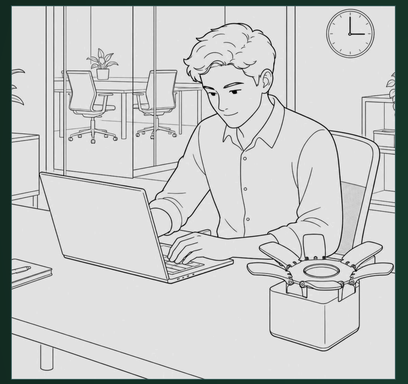
Sketch Development | 草图进展



System Flow | 系统流程图



Storyboard | 故事板



The user sat down and began to focus on working at the desk.
用户坐下后，开始专心在办公桌前工作。



As screen work continues, sedentary time accumulates unnoticed.
随着屏幕时间增加，久坐时间不知不觉累积。



Calmpa's petals close slowly, making the sedentary state visible.
Calmpa花瓣缓缓闭合，使静止状态显露出来。



Subtle changes remind users to pay attention to their health.
细微的变化会提醒用户关注自身健康。



Users can leave their seats for a short rest to allow their bodies to recover.
用户可以离开座位稍作休息，让身体恢复。



Calmpa slowly reopens, symbolizing physical recovery.
Calmpa 缓缓重新开放，象征着身体的恢复。

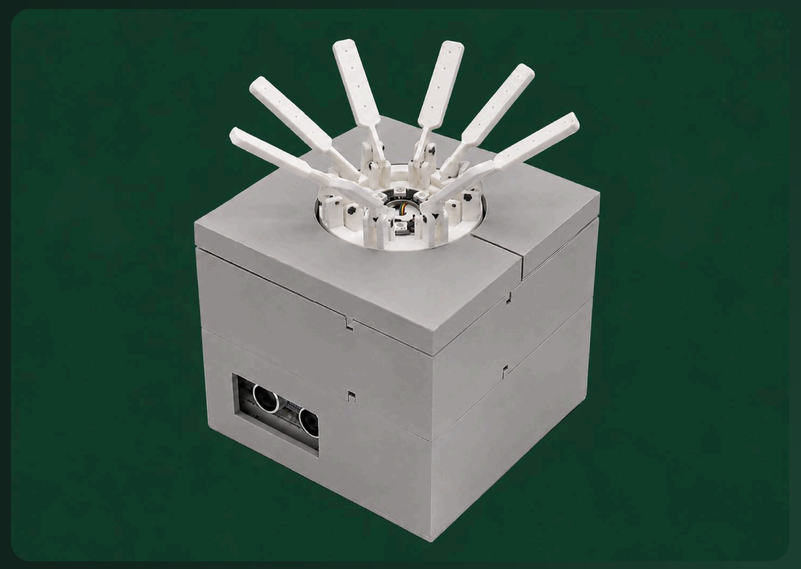
CALMPA

Design of a Non-Intrusive Desktop Interactive Device for Break Reminders During Prolonged Desk Work

设计一种非侵入式桌面交互设备，用于在长时间伏案工作期间提醒休息时间

From concept to working prototype, Calmpa is realized as an interactive device that quietly responds to presence and encourages natural breaks through subtle physical expression.

从概念到工作原型，Calmpa 被设计成一款交互式设备，它能安静地回应人们的存在，并通过微妙的身体表达来鼓励自然的休息





Calm Interaction
平静交互

Gentle response that respects concentration.

温和的回应，尊重对方的专注力

Embedded Reminder
嵌入式提醒

Make rest visible through physical movement.

通过物理动作让休息变得可见

Seamless Presence
低违和感

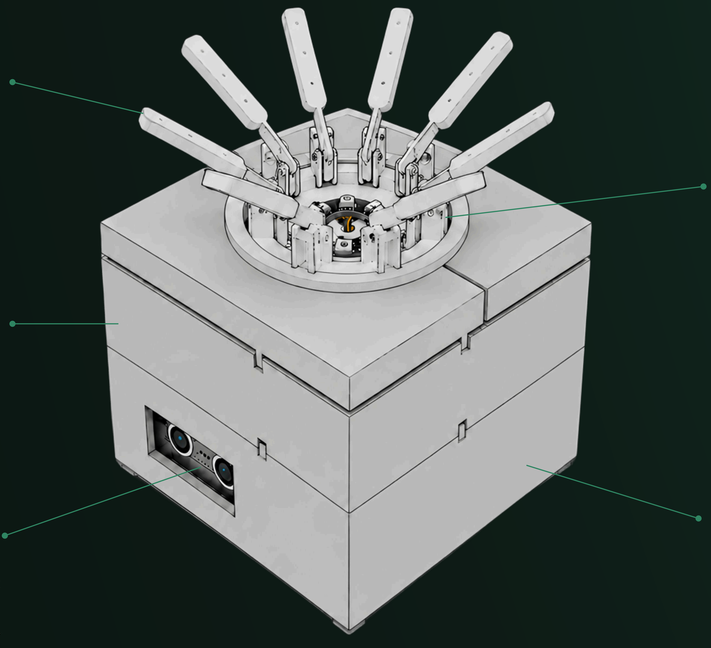
Minimal form that blends into dally workspace.

完美融入日常工作空间

Prototype Introduction | 原型介绍

Prototype Goal 原型目标

- Provide calm, non-intrusive break reminders
  提供平静、非侵入式的休息提醒
- Express state changes through subtle motion & light
  通过微妙的动作与光线传达状态
- Integrate naturally on the desktop
  自然融入办公桌面环境



Mechanical Petal Arms
机械花瓣臂

The hinged radial arms open and close together, forming the main physical feedback.

铰链式径向臂一起打开和关闭，形成主要的物理反馈

Cubic Housing
立体部分

The compact cubic shell stabilizes and protects the components.

紧凑的立方体外壳能够稳定和保护内部组件

Front Sensing Module
前部感应模块

The front ultrasonic sensor detects desk presence.

前置超声波传感器可检测桌面上是否有人

Circular Top Opening
圆形顶部开口

The circular opening supports smooth radial movement.

圆形开口支持平稳的径向运动

Layered Structure / Internal Components
分层结构 / 内部组件

The layered housing accommodates mechanisms, electronics, and power management.

多层外壳可容纳机械装置、电子元件和电源管理部件

Prototype Making 原型制作



Ultrasonic Sensor
超声波传感器

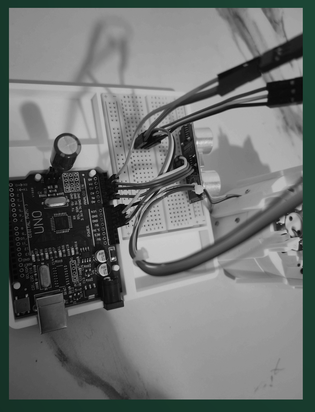


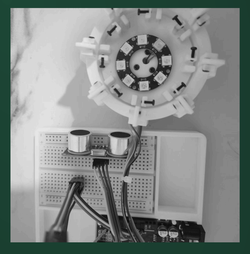
Servo Motors
伺服电机

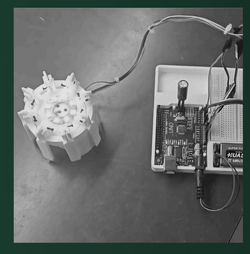


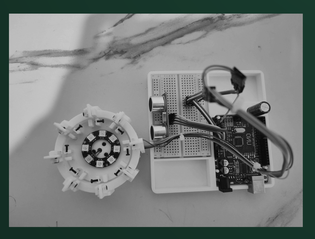
Arduino Board
开发板

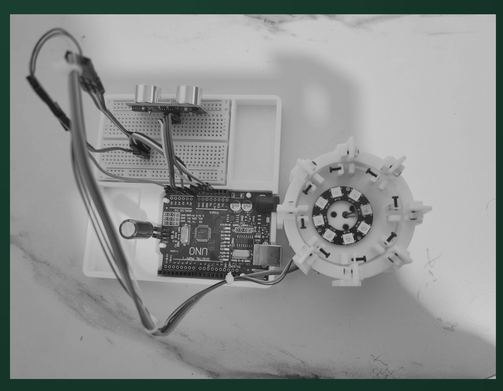
Making Process 制作过程









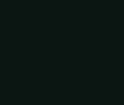


Prototype Test | 原型测试

The prototype was tested to verify its sensing, motion, and feedback performance in a desk work environment.

对该原型进行了测试，以验证其在办公桌工作场景中的传感、运动和反馈性能

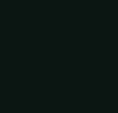
① Presence Detection 存在性检测



Check if the ultrasonic sensor can detect user presence within range.

验证有效范围内的用户检测

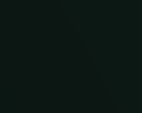
② Petal Movement 花瓣移动



Test whether the servo motor can smoothly drive the petals.

验证舵机能否平稳驱动花瓣运动

③ Calm Feedback 平静反馈



Observe if the movement feels slow, steady, and non-intrusive.

观察反馈是否缓慢、稳定且非打扰

Method & Notes 测试方式与说明

The prototype was placed on a desk and observed under different user distances and sitting durations. The test focused on sensing response, motion quality, and feedback clarity.

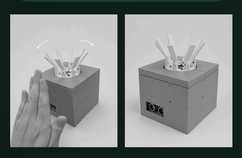
在桌面环境中测试不同距离与久坐时长下的感知、运动与反馈表现

Functional Test | 功能测试

The prototype provides calm feedback through gentle opening and closing movements.

原型通过温和的开合动作提供平静反馈

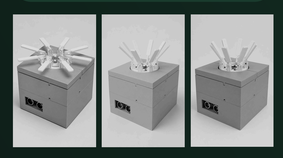
Sensor Response Test 传感器响应测试



The ultrasonic sensor was tested at different distances. It was able to detect the presence of the table within its actual effective range and send a signal to the mobile system.

传感器在不同距离进行了测试。它能够在有效范围内检测用户是否在桌前，并向系统发送信号

Petal Movement Test 花瓣运动测试



The servo motor successfully drove the mechanical petal arm to switch between opening and closing states. This confirms that prolonged sitting can be translated into visible body movement.

伺服电机成功驱动机械花瓣臂在开合状态之间切换。这证实了长时间久坐可以转化为可见的身体运动

Motion Quality Observation 运动质量观察



Different movement speeds were observed during the test. Faster movement felt more abrupt, while slower movement felt calmer,more stable, and less intrusive.

测试过程中观察到不同的运动速度。较快的运动感觉更突兀，而较慢的运动则感觉更平静、更稳定、更不突兀

Motion Test | 动态测试

Motion testing adjusts servo angle, speed, and position to ensure stable movement.

运动测试会调整伺服角度、速度和位置，以确保运动稳定

Open State 开启状态



Closed State 闭合状态



Speed Level	速度水平	Movement Effect	运动效果
Fast	快速	Too fast, abrupt, and distracting.	太快、太突然、太分散注意力
Medium	中速	It's acceptable, but still obvious.	这可以接受，但仍然很明显
Slow	慢速	Calm, smooth and non-intrusive.	平静、柔和、并且不突兀

Design Value | 设计价值

Calmpa's ultimate design delivers tangible value by providing calm, non-invasive support in theoffice environment, helping people improve their awareness of sedentary lifestyles.

Calmpa 的终极设计通过在办公环境中提供平静、非侵入式的支持来带来切实价值，帮助人们提高对久坐生活方式的认识

Non-intrusive Reminder
非侵入式提醒

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Visible Sedentary Awareness
可视化久坐意识

The petal motion offers a clear and intuitive cue for prolonged sitting, helping users become more mindful of their posture and movement.

花瓣的运动为长时间坐着提供了清晰直观的提示，帮助用户更加注意自己的姿势和动作

Calm Office Support
温和办公支持

The product features a simple design and smooth operation, allowing it to blend seamlessly into the office space and enhance employee well-being without drawing attention.

该产品设计简洁，操作流畅，能够无缝融入办公空间，在不引起注意的情况下提升员工福祉

Functional Prototype to Product Vision | 从功能原型到产品化愿景



→



The physical prototype proves how Calmpa works, while the refined visualization shows how it could evolve.

实体原型证明 Calmpa 如何运作，而产品化效果图展示它未来可以如何发展

The functional prototype focuses on validating Calmpa's core interaction logic: desk presence detection, servo-driven petal movement, and calm physical feedback. Its visible structure, sensor opening, and exposed mechanical parts reflect the current stage of prototyping, where functionality and feasibility are the main priorities.

功能原型主要用于验证 Calmpa 的核心交互逻辑：桌前状态检测、舵机驱动花瓣运动，以及温和的物理反馈。其中可见的结构、传感器开口和外露的机械部件，反映了当前原型阶段以功能可行性和机制验证为重点

The refined product vision builds upon this working prototype. It explores how Calmpa could become a more integrated, compact, and office-friendly desktop device through smoother surfaces, hidden components, softer materials, and a cleaner visual language.

产品化愿景则基于该功能原型进一步发展，探索 Calmpa 如何通过更完整的外壳、更隐藏的组件、更柔和的材料和更简洁的视觉语言，转化为一个更一体化、更适合日常办公环境的桌面装置